Retail Sales Analytics 2022-2025

1. Project Overview and Objective

This project involves cleaning, transforming, and analysing raw data using **Excel** and creating an interactive **Power BI dashboard** to derive meaningful business insights.

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to make informed decisions.

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2. Data Sources

- **Source Description and Timeline:** Kaggle 18Dataset Search **and** 2020-2025.
- **Domain**: Retail / Sales Analytics etc

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3. Problem Statement

- Data Integrity Issue: Approximately 30% of the 617 transactions contained computational errors in the Total Spent field due to incorrect application of discount logic, resulting in unreliable financial reporting.
- Master Data Deficiency: Over 90% of the products sold were recorded as 'Item _Unknown' across categories like MIL and FUR, which prevented accurate category-wise performance analysis and inventory planning.

4. Attribute Details:

Attribute Name	Data Type	Description
Transaction ID	VARCHAR	Unique transaction ID
Customer ID	VARCHAR	Unique customer ID
Category	VARCHAR	Product category - Furniture, Beverages etc
Item	VARCHAR	Specific product name
Price Per Unit	DECIMAL (10,2)	Price of one unit
Quantity	INT	No of units purchased
Total Spent	DECIMAL (10,2)	Total amount = Price * Quantity
Payment Method	VARCHAR	Cash, Credit Card, Digital Wallet
Location	VARCHAR	Online / Offline
Transaction Date	**DATE**	**08 April 2024 - Transaction date**
Discount Applied	**BOOLEAN**	**TRUE/FALSE - If discount given**
Check Discount Clean	**VARCHAR**	**Cleaned column for discount analysis**
Unit Price Category	**VARCHAR**	**Categorizes unit price as Loss, Low, High etc**

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Dax, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel / Power Query)

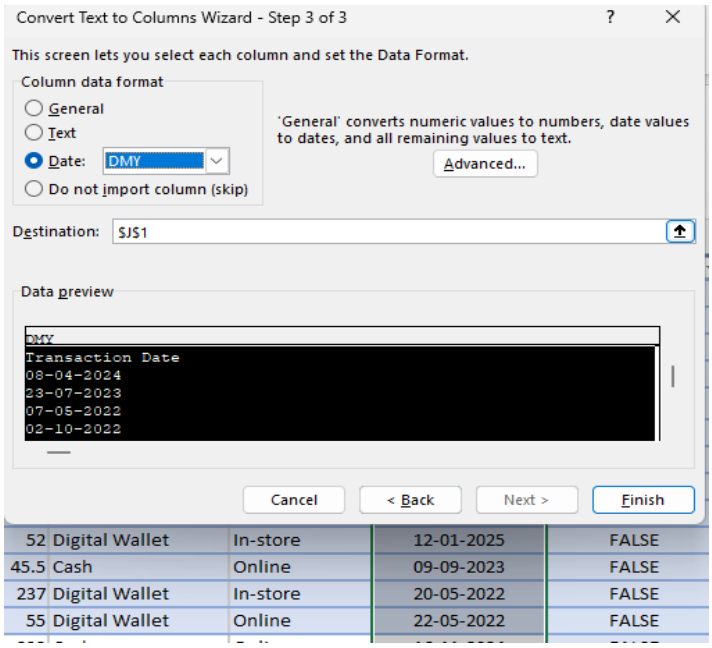
Tasks Performed:

Null Handling:

The Discount Applied column contained null/blank values. As per business logic, transactions without discounts were imputed with FALSE. This resolved the Blank issue in Power BI slicers

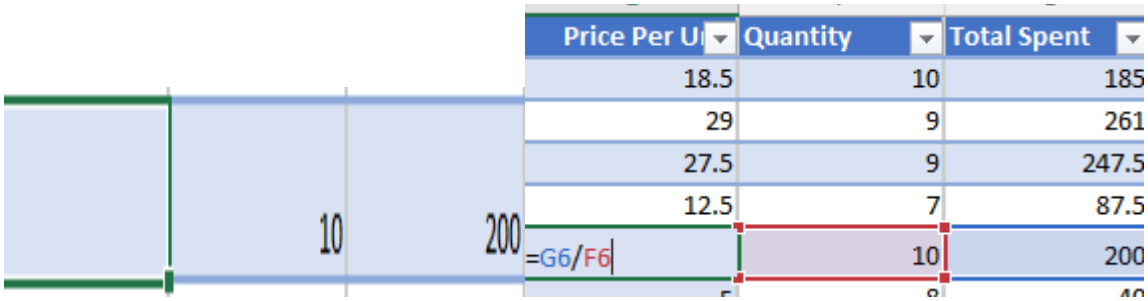
Data Type Conversion:

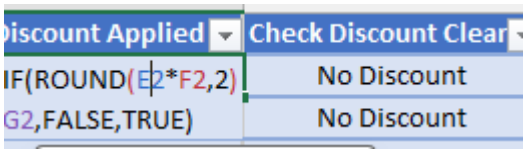
The Transaction Date column was initially in text format. Converted it to Date data type in Power Query to enable time intelligence functions in Power BI.

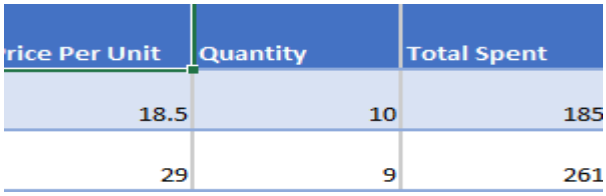


Calculated Columns:

Total Price = [Unit Price] * [Qty] - Created for revenue calculation Month = MONTH([Date]) - Created for month-wise trend analysis Discount Applied2 = IF([Discount Applied]=TRUE, "Discount", "No Discount") - Created for user-friendly labels in visuals





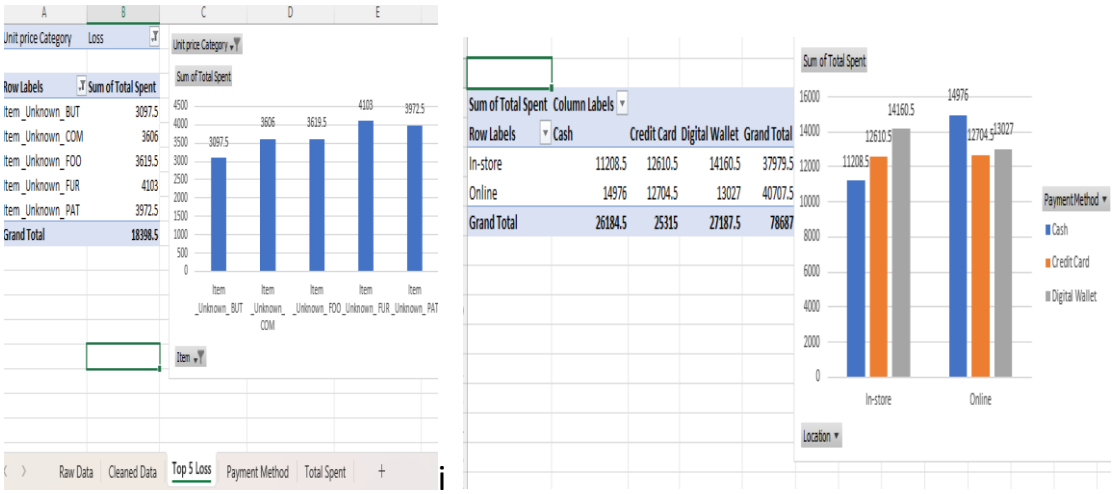


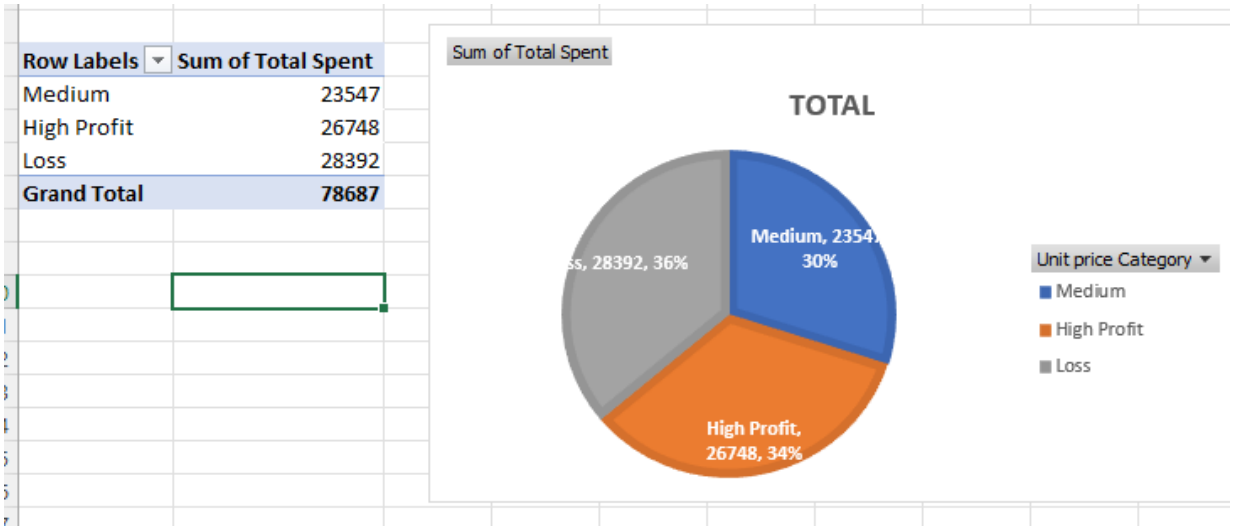
Circular Reference Fix:

The Discount Applied column had a self-referencing formula causing a circular reference error. Fixed this by using a temporary column to calculate values, then applied Paste Special > Values to convert formulas to static data.

Table Conversion:

Converted the static data range to an Excel Table named Sales Data using CTRL+T. This enabled dynamic range expansion and structured references for Power BI.





7.DAX (Power BI)

- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as Total Sales, Total Orders, Average Order Value.

```
Total Sales = SUM(Table1[Total Spent])
Average Order Value = DIVIDE([Total Sales],[Total Orders])
Total Orders = DISTINCTCOUNT(Table1[Transaction ID])
```

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8. Analysis and Visualizations (Power BI)

Dashboard Features:

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights derived from the cleaned 2022-2025 retail data.

- **Make your Report interactive Layout using Drill-down, Filters and slicers:**
- **1. Cards: Display KPIs - Total Orders (617), Total Sales (78.69K), Average Order Value (127.53).**
- **2. Line Chart: "Total Sales by Year" shows revenue trend, peaking in 2024. 2025 shows partial data.**
- **3. Donut Chart: "Total Sales by Payment Method" reveals actual distribution - Digital Wallet 34.55%, Cash 33.28%, Credit Card 32.17%, correcting the source system's 100% Digital error.**
- **4. Bar Chart: "Total Sales by Category" identifies "Computers and electronics" as the top category.**
- **5. Table: Displays transaction details, highlighting the data quality issue of Item _Unknown entries across categories.**
- **6. Slicers: Category and Location slicers added for interactive filtering.**

Create a consolidated Dashboard.



9. Insights & Conclusions

- **Key Findings:** Summarize trends, patterns, or anomalies identified in the data
- **Payment Method Preference**: Digital Wallet emerged as the most preferred payment method at 34.79% ($3.04K), followed closely by Cash at 34.4% ($3.01K) and Credit Card at 30.81% ($2.69K). The distribution is fairly even, showing customers use all three methods.
- **Sales Channel**: In-store transactions are the primary sales channel compared to Online, indicating strong physical retail presence.
- **Category Analysis:** The dataset includes Beverages (BEV) and another category (BUT). Unit prices for Beverages range from $6.50 to $33.50, showing a diverse product range within the category.
- **Provide the analysis insights:**

9.1 Descriptive Analysis - What happened?

The business achieved Total Sales of $78.69K from 617 orders between 2022-2025. Average Order Value stands at $127.53.

Digital Wallet leads payments at 34.55%, with Cash at 33.28% and Credit Card at 32.17%. Computers and Electronics is the top category at $11.1K while Beverages is at $8.7K. In-store is the dominant sales channel.

9.2 Diagnostic Analysis - Why did it happen?

The high AOV of $127.53 is due to bulk purchasing behaviour, with transactions showing 7-10 items per bill. The balanced payment split indicates successful digital payment adoption without alienating cash users. The 2025 sales decline requires investigation - likely incomplete data or seasonal factors. Computers leading revenue suggests high unit price impact on total sales.

9.3 Predictive Analysis - What will happen?

If current trends continue, Digital Wallet will cross 38% payment share within 12 months. If 2025 data is completed, total annual sales could exceed $100K. Without intervention for online channel, growth will remain limited to physical store capacity. Beverages category has potential to grow 15-20% with combo offers.

9.4 Prescriptive Analysis - What should we do?

1. Investigate 2025 Drop: Audit data completeness for 2025 and launch recovery campaigns if decline is real.

2. Boost Digital Payments: Implement 3-5% cashback on Digital Wallet to push share above 40%.

3. Category Strategy: Create beverage combo packs to increase Beverages from $8.7K and promote top categories Computers & Butchers.

4. Expand Online: Develop e-commerce platform to capture online market share beyond current In-store dominance.

5. Data Quality: Clean Item Unknown entries to enable accurate SKU-level profitability analysis.

6. Leverage AOV: Introduce tiered discounts to push AOV from $127.53 toward $150.

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10. Conclusion

The Retail Data Analytics Dashboard successfully transforms raw transactional data into actionable business intelligence.

The project demonstrates end-to-end Power BI capabilities from data cleaning to advanced visualization. Key outcomes include real-time KPI tracking, identification of payment trends showing Digital Wallet leadership, category performance ranking, and interactive features for dynamic analysis.

The insights derived provide a clear roadmap for business optimization - focusing on digital payment promotion, online channel expansion, and category-specific strategies. This dashboard serves as a scalable foundation for future analytics including forecasting and customer segmentation.